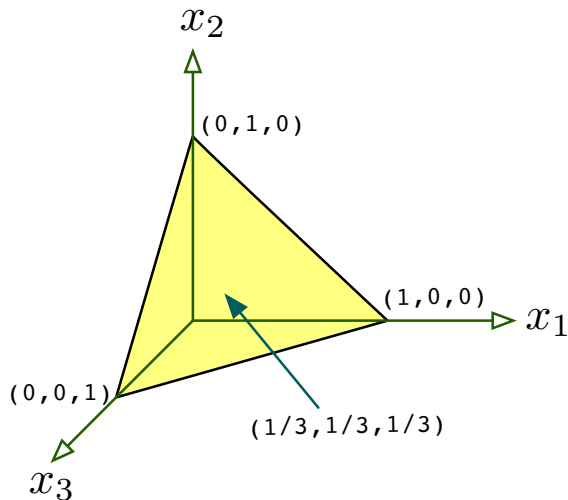


Solutions to HW 1

1. (a) $\mathcal{X} = \mathbb{R}^{10}$.
 (b) $\mathcal{X} = [0, 1]^3$.
2. (a) 9
 (b) $\sqrt{7^2 + 24^2} = 25$
 (c) $\sqrt{4^2 + 3^2 + 12^2} = 13$
3. (a) $p = \begin{bmatrix} 1/5 \\ 3/10 \\ 1/2 \end{bmatrix}$
 (b) Δ_3
4. Any point with at least one non-negative coordinate, e.g. $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.
5. Sketch of probability simplex Δ_3 :



6. (a) $\|p - q\|_1 = (1/4) + (0) + (1/8) + (1/8) = 1/2$
 (b) $\|q - r\|_1 = (1/4) + (1/4) + (0) + (0) = 1/2$
 (c) $K(p, q) = (1/2) \log 2 + (1/4) \log 1 + (1/8) \log(1/2) + (1/8) \log 2 = 1/4 \log(2)$
 (d) $K(q, r) = (1/2) \log(1/2) + (1/4) \log \infty + (1/4) \log 1 + (1/4) \log 1 = \infty$
7. *Choosing representations for nearest neighbor.* See solution notebook.

(a,b) Dimensionalities and accuracies of the representations:

Representation	Dimension	1-NN accuracy
Raw pixel	3072	35.4%
HoG	512	36.6%
VGG-last-fc	4096	92.1%
VGG-last-conv	512	92.0%
random-VGG-last-fc	4096	40.4%
random-VGG-last-conv	512	38.9%

(c) See solution notebook.

8. *Word vectors*. Here are the nearest neighbors:

- communism: ['fascism', 'capitalism', 'nazism', 'stalinism', 'socialism']
- africa: ['african', 'continent', 'south', 'africans', 'zimbabwe']
- happy: ['glad', 'pleased', 'always', 'everyone', 'sure']
- sad: ['sorry', 'tragic', 'happy', 'pathetic', 'awful']
- upset: ['upsetting', 'surprised', 'upsets', 'stunned', 'shocked']
- computer: ['computers', 'software', 'technology', 'laptop', 'computing']
- cat: ['cats', 'dog', 'pet', 'feline', 'dogs']
- dollar: ['currency', 'dollars', 'euro', 'multibillion', 'weaker']